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1. We say a graph $G = (V, E)$ is *topologically disconnected* if we can find two sets $U, W \subset V$ such that:

1. $U \cup W = V$,
2. $U \cap W = \emptyset$,
3. Both U and W are non-empty, and
4. There are no edges in G connecting a vertex in U to a vertex in W .

(The phrase “topologically connected” is *not* standard mathematical terminology.) Prove that a graph is connected if and only if it is not topologically disconnected.

Since the question of whether the empty graph or the graph with one vertex are connected is a matter more for philosophy than mathematics, you may ignore these cases and consider only graphs with two or more vertices. This should not seriously impact your proof.

Hint: Proof by contradiction.

Solution: Assume that G is topologically connected (that is to say, not topologically disconnected) and assume that there exist two vertices x and y with no paths between them. Let U be the set of all vertices with paths to x and let W be the set of all vertices with paths to y . If there are any vertices left over, put them in U . Then $U \cup W = V$ by construction. If there were a connection from a vertex in U to a vertex in W , then that would provide a path from y to either x – which is impossible – or to one of the other vertices we’ve added to U , which we previously stated had no path to x .

Now assume that G is connected and take two sets U and W such that $U \cup W = V$, $U \cap W = \emptyset$, and neither U nor W are empty. Then we can take a vertex in U and a vertex in W . There exists a path between these vertices, which must contain an edge connecting a vertex in U to a vertex in W .

2. Let $G = (V, E)$ be a graph. A set of vertices $I \subseteq V$ is called an *independent set* if there are no edges between any pair of vertices in I . Namely, $x, y \in I$ implies $(x, y) \notin E$. Let C be the size of the largest independent set in G . Prove the chromatic number $\chi(G)$ satisfies

$$\chi(G) \geq \frac{|V|}{C}.$$

Hint below if you are stuck.¹

¹Assume you have a k -coloring. How can you use this to build an independent set?

Solution: Suppose that $\chi(G) = k$. Take any k -coloring and let V_i be the vertices that are colored color i for $i = 1, \dots, k$. Then each V_i is an independent set by definition of a coloring. Hence $|V_i| \leq C$ as C is the size of the largest independent set. Thus:

$$|V| = \sum_{i=1}^k |V_i| \leq Ck$$

Hence $k \geq \frac{|V|}{C}$.